```
Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]> create labwork
    -> ;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MariaDB server version for the right syntax to use near 'labwork' at line
MariaDB [(none)]> CREATE DATABASE labwork;
Query OK, 1 row affected (0.005 sec)

MariaDB [(none)]> USE labwork;
Database changed
MariaDB [labwork]>
MariaDB [labwork]> CREATE TABLE salesman (
    ->     salesman_id INT PRIMARY KEY,
    ->     name VARCHAR(100) NOT NULL,
    ->     city VARCHAR(100) NOT NULL,
    ->     commission DECIMAL(4,2) NOT NULL CHECK (commission >= 0 AND commission <= 1)
    -> );
Query OK, 0 rows affected (0.025 sec)

MariaDB [labwork]>
MariaDB [labwork]> CREATE TABLE customer (
    ->     customer_id INT PRIMARY KEY,
    ->     customer_name VARCHAR(100) NOT NULL,
    ->     city VARCHAR(100) NOT NULL,
    ->     grade INT CHECK (grade BETWEEN 100 AND 500),
    ->     salesman_id INT,
    ->     FOREIGN KEY (salesman_id) REFERENCES salesman(salesman_id)
    -> );
Query OK, 0 rows affected (0.017 sec)

MariaDB [labwork]>
MariaDB [labwork]> CREATE TABLE orders (
    ->     order_no INT PRIMARY KEY,
    ->     purchase_amt DECIMAL(10,2) NOT NULL,
    ->     order_date DATE NOT NULL,
    ->     customer_id INT NOT NULL,
    ->     salesman_id INT NOT NULL,
    ->     FOREIGN KEY (customer_id) REFERENCES customer(customer_id),
    ->     FOREIGN KEY (salesman_id) REFERENCES salesman(salesman_id)
    -> );
Query OK, 0 rows affected (0.025 sec)

MariaDB [labwork]>
```

```
ariaDB [labwork]> CREATE TABLE orders (
    ->     order_no INT PRIMARY KEY,
    ->     purchase_amt DECIMAL(10,2) NOT NULL,
    ->     order_date DATE NOT NULL,
    ->     customer_id INT NOT NULL,
    ->     salesman_id INT NOT NULL,
    ->     FOREIGN KEY (customer_id) REFERENCES customer(customer_id),
    ->     FOREIGN KEY (salesman_id) REFERENCES salesman(salesman_id)
    -> );
uery OK, 0 rows affected (0.025 sec)

ariaDB [labwork]> INSERT INTO salesman (salesman_id, name, city, commission) VALUES
    -> (101, 'John Smith', 'New York', 0.15),
    -> (102, 'Sara Khan', 'Chicago', 0.12),
    -> (103, 'David Lee', 'Los Angeles', 0.10);
uery OK, 3 rows affected (0.032 sec)
ecords: 3  Duplicates: 0  Warnings: 0

ariaDB [labwork]>
ariaDB [labwork]> INSERT INTO customer (customer_id, customer_name, city, grade, sales
an_id) VALUES
    -> (201, 'Alice Brown', 'New York', 300, 101),
    -> (202, 'Bob Martin', 'Chicago', 200, 102),
    -> (203, 'Carol White', 'Los Angeles', 400, 103),
    -> (204, 'Daniel Green', 'Boston', 150, 101);
uery OK, 4 rows affected (0.004 sec)
ecords: 4  Duplicates: 0  Warnings: 0

ariaDB [labwork]>
ariaDB [labwork]> INSERT INTO orders (order_no, purchase_amt, order_date, customer_id,
salesman_id) VALUES
    -> (1001, 2500.00, '2025-03-01', 201, 101),
    -> (1002, 1800.50, '2025-03-02', 202, 102),
    -> (1003, 3200.75, '2025-03-03', 203, 103),
    -> (1004, 950.00,  '2025-03-04', 204, 101),
    -> (1005, 4100.00, '2025-03-05', 201, 101);
uery OK, 5 rows affected (0.009 sec)
ecords: 5  Duplicates: 0  Warnings: 0

ariaDB [labwork]>
```

```java
public class DBConnection {
    public static void main(String[] args) {
        if (args.length != 3) {
            System.out.println("Usage: java DBConnection <username> <password> <database>");
            return;
        }

        String username = args[0];
        String password = args[1];
        String database = args[2];

        String url = "jdbc:mariadb://localhost:3306/" + database;

        ArrayList<Sales> salesList = new ArrayList<>();

        String query =
                "SELECT o.order_no, c.customer_name, c.city AS customer_city, " +
                "       s.name AS salesman_name, o.purchase_amt, " +
                "       (o.purchase_amt * s.commission) AS commission_amount " +
                "FROM orders o " +
                "JOIN customer c ON o.customer_id = c.customer_id " +
                "JOIN salesman s ON o.salesman_id = s.salesman_id " +
                "ORDER BY o.order_no";

        try {
            Class.forName("org.mariadb.jdbc.Driver");

            try (Connection conn = DriverManager.getConnection(url, username, password);
                 PreparedStatement pstmt = conn.prepareStatement(query);
                 ResultSet rs = pstmt.executeQuery()) {

                while (rs.next()) {
                    Sales sale = new Sales(
                            orderNumber: rs.getInt("order_no"),
                            customerName: rs.getString("customer_name"),
                            customerCity: rs.getString("customer_city"),
                            salesmanName: rs.getString("salesman_name"),
                            amount: rs.getDouble("purchase_amt"),
                            commissionAmount: rs.getDouble("commission_amount")
                    );

                    salesList.add(sale);
                }
```

```
            }

            System.out.println("Connected successfully to database: " + database);
            System.out.println("Sales records retrieved:\n");

            for (Sales sale : salesList) {
                System.out.println(sale);
            }
            rs.close();
            pstmt.close();
            conn.close();
        }

    } catch (ClassNotFoundException e) {
        System.out.println("MariaDB JDBC Driver not found.");
        e.printStackTrace();
    } catch (SQLException e) {
        System.out.println("Database connection/query error.");
        e.printStackTrace();
    }
  }
}
```

```java
package week7;

import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.PreparedStatement;
import java.sql.ResultSet;
import java.sql.SQLException;
import java.util.ArrayList;

class Sales {
    int orderNumber;
    String customerName;
    String customerCity;
    String salesmanName;
    double amount;
    double commissionAmount;

    public Sales(int orderNumber, String customerName, String customerCity,
                 String salesmanName, double amount, double commissionAmount) {
        this.orderNumber = orderNumber;
        this.customerName = customerName;
        this.customerCity = customerCity;
        this.salesmanName = salesmanName;
        this.amount = amount;
        this.commissionAmount = commissionAmount;
    }

    @Override
    public String toString() {
        return "Sales{" +
                "orderNumber=" + orderNumber +
                ", customerName='" + customerName + '\'' +
                ", customerCity='" + customerCity + '\'' +
                ", salesmanName='" + salesmanName + '\'' +
                ", amount=" + amount +
                ", commissionAmount=" + commissionAmount +
                '}';
    }
}
```

```java
package week7;

import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.PreparedStatement;
import java.sql.ResultSet;
import java.sql.SQLException;
import java.util.ArrayList;

class Sales {
    int orderNumber;
    String customerName;
    String customerCity;
    String salesmanName;
    double amount;
    double commissionAmount;

    public Sales(int orderNumber, String customerName, String customerCity,
                 String salesmanName, double amount, double commissionAmount) {
        this.orderNumber = orderNumber;
        this.customerName = customerName;
        this.customerCity = customerCity;
        this.salesmanName = salesmanName;
        this.amount = amount;
        this.commissionAmount = commissionAmount;
    }

    @Override
    public String toString() {
        return "Sales{" +
                "orderNumber=" + orderNumber +
                ", customerName='" + customerName + '\'' +
                ", customerCity='" + customerCity + '\'' +
                ", salesmanName='" + salesmanName + '\'' +
                ", amount=" + amount +
                ", commissionAmount=" + commissionAmount +
                '}';
    }
}
```

```
Connected successfully to database: labwork
Sales records retrieved:

Sales{orderNumber=1001, customerName='Alice Brown', customerCity='New York', salesmanName='John Smith', amount=2500.0, commissionAmount=375.0}
Sales{orderNumber=1002, customerName='Bob Martin', customerCity='Chicago', salesmanName='Sara Khan', amount=1800.5, commissionAmount=216.06}
Sales{orderNumber=1003, customerName='Carol White', customerCity='Los Angeles', salesmanName='David Lee', amount=3200.75, commissionAmount=320.075}
Sales{orderNumber=1004, customerName='Daniel Green', customerCity='Boston', salesmanName='John Smith', amount=950.0, commissionAmount=142.5}
Sales{orderNumber=1005, customerName='Alice Brown', customerCity='New York', salesmanName='John Smith', amount=4100.0, commissionAmount=615.0}
```